

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I **P.Paneendra (192321072)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:P.Paneendra

Annexure A

Analytical Problem Solving

[1] RSA Encryption Analysis

Given $p=11$, $q=13$

[A] Compute n and $\phi(n)$

[B] Choose $e=7$ and find the private key d

[C] Encrypt the message $M=9$

[D] Decrypt the ciphertext and verify the result

[E] Analyze how changing e affects security

ANS: Analytical Problem Solving

1. RSA Encryption Analysis

Given:

- $p = 11$
- $q = 13$
- $e = 7$
- Message $M = 9$

(a) Compute n and $\phi(n)$

$$n = p \times q = 11 \times 13 = 143$$

$$\phi(n) = (p - 1)(q - 1) = 10 \times 12 = 120$$

Answer:

- $n = 143$
- $\phi(n) = 120$

(b) Find the private key d

Find d such that

$$7d \equiv 1 \pmod{120}$$

Using the Extended Euclidean Algorithm,

$$d = 103$$

(Check: $7 \times 103 = 721$, $721 \bmod 120 = 1$)

Private Key:

$$d = 103$$

(c) Encrypt the message

Formula:

$$\begin{aligned} C &= M^e \bmod n \\ C &= 9^7 \bmod 143 \end{aligned}$$

Using modular arithmetic,

$$9^7 \bmod 143 = 48$$

Ciphertext:

$$\boxed{C = 48}$$

(d) Decrypt the ciphertext

$$\begin{aligned} M &= C^d \bmod n \\ M &= 48^{103} \bmod 143 \end{aligned}$$

Result:

$$\boxed{M = 9}$$

The decrypted message matches the original plaintext.

(e) Effect of changing e

- Smaller e (e.g., 3) $\rightarrow$ Faster encryption but may introduce security risks if used improperly.
- Larger $e \rightarrow$ Slower encryption but generally safer.
- **Common choice:** $e = 65537$ because it provides a good balance of security and efficiency.

[2] Block Cipher Mode Evaluation

Given plaintext blocks:

$P = [1010, 1010, 1111, 1010]$, Key = $K = 1100$

[A] Encrypt using ECB and CBC modes (Assume IV = 0000)

[B] Compare the ciphertext outputs

[C] Analyze which mode hides patterns better and explain why

ANS: Block Cipher Mode Evaluation

Given:

Plaintext blocks

$P_1 = 1010$

$P_2 = 1010$

$P_3 = 1111$

$P_4 = 1010$

Key

$K = 1100$

IV

0000

Assume encryption uses XOR:

$$C = P \oplus K$$

(a) ECB Encryption

Block 1

1010

1100

0110

C1 = **0110**

Block 2

Same plaintext

C2 = **0110**

Block 3

1111

1100

0011

C3 = **0011**

Block 4

Again

C4 = **0110**

ECB Ciphertext

[0110, 0110, 0011, 0110]

CBC Encryption

Formula

$$C_i = (P_i \oplus C_{i-1}) \oplus K$$

IV = 0000

Block 1

$$1010 \oplus 0000 = 1010$$

$$1010 \oplus 1100 = 0110$$

$$C1 = \mathbf{0110}$$

Block 2

$$1010 \oplus 0110 = 1100$$

$$1100 \oplus 1100 = 0000$$

$$C2 = \mathbf{0000}$$

Block 3

$$1111 \oplus 0000 = 1111$$

$$1111 \oplus 1100 = 0011$$

$$C3 = \mathbf{0011}$$

Block 4

$$1010 \oplus 0011 = 1001$$

$$1001 \oplus 1100 = 0101$$

$$C4 = \mathbf{0101}$$

CBC Ciphertext

[0110,0000,0011,0101]

(b) Comparison

ECB

0110 0110 0011 0110

CBC

0110 0000 0011 0101

Repeated plaintext blocks produce repeated ciphertext in ECB but not in CBC.

(c) Better mode

CBC is better because:

- Uses an Initialization Vector (IV).
- Each ciphertext block depends on the previous ciphertext block.
- Hides repeating patterns.

[3] DES vs 3-DES vs Blowfish Performance Study

Given:

[A] DES key size = 56 bits, time = 2 ms/block

[B] 3-DES key size = 168 bits, time = 6 ms/block

[C] Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

- Calculate total encryption time for each algorithm**
- Analyze trade-offs between security and performance**
- Recommend the best algorithm for secure real-time communication**

ANS: DES vs 3-DES vs Blowfish Performance Study

Given

Algorithm Time/block

DES 2 ms

3DES 6 ms

Blowfish 1 ms

Blocks = **1000**

(i) Total Encryption Time

DES

$$1000 \times 2 = 2000 \text{ ms} = 2 \text{ s}$$

3DES

$$1000 \times 6 = 6000 \text{ ms} = 6 \text{ s}$$

Blowfish

$$1000 \times 1 = 1000 \text{ ms} = 1 \text{ s}$$

Answer

Algorithm Total Time

DES 2 s

3DES 6 s

Blowfish 1 s

(ii) Trade-offs

Algorithm Security Performance

DES Low Medium

3DES High Slow

Blowfish High Fast

(iii) Recommendation

Blowfish

Reason:

- Fastest encryption
- Strong security
- Suitable for real-time communication

[4] Diffie-Hellman Key Exchange Analysis

Given:

[A] $p=23, g=5$

[B] User A private key = 6

[C] User B private key = 15

[D] Compute public keys for A and B

[E] Compute the shared secret key

[F] Demonstrate how a Man-in-the-Middle attacker could alter the exchange

[G] Suggest a secure method to prevent the attack

ANS: Diffie-Hellman Key Exchange

Given

- $p = 23$
- $g = 5$
- A private key = 6
- B private key = 15

(d) Public Keys

User A

$$A = 5^6 \bmod 23$$
$$15625 \bmod 23 = 8$$

User B

$$B = 5^{15} \bmod 23$$
$$= 19$$

Public Keys

- $A = 8$
- $B = 19$

(e) Shared Secret

User A

$$19^6 \bmod 23 = 2$$

User B

$$8^{15} \bmod 23 = 2$$

Shared secret

2

(f) Man-in-the-Middle Attack

Attacker intercepts public keys.

Instead of

$A \rightarrow B : 8$

attacker sends

12

Instead of

$B \rightarrow A : 19$

attacker sends

15

Now

- A shares a key with attacker.
 - B shares another key with attacker.
 - Attacker decrypts and modifies all messages.
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(g) Prevention

Use

- Digital certificates

- Digital signatures
- Authenticated Diffie-Hellman
- TLS

These verify the identity of both users and prevent MITM attacks.

[5] ECC vs RSA Comparison

Given:

[A] RSA key size = 1024 bits

[B] ECC key size = 160 bits (equivalent security)

[C] Device processing capacity = limited (IoT device)

[D] Analyze computational cost for both systems

[E] Estimate which consumes less power and memory

[F] Justify which cryptosystem is better for the given scenario with reasoning

ANS: ECC vs RSA Comparison

Given

- RSA = 1024-bit
- ECC = 160-bit
- IoT device with limited resources

(d) Computational Cost

RSA

- Large keys
- More CPU operations
- Slower

ECC

- Small keys
- Faster calculations
- Lower CPU usage

ECC has lower computational cost.

(e) Power and Memory

Feature	RSA	ECC
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Power	High	Low
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Memory	High	Low
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Key Size	1024-bit	160-bit
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ECC consumes less power and memory.

(f) Best Choice

ECC is the better choice because:

- Provides security equivalent to RSA with much smaller keys.
- Requires less computation.
- Consumes less power and memory.
- Is ideal for resource-constrained IoT devices.